Journal of Rare Earths 37 (2019) 1040–1046

Contents lists available at ScienceDirect

Journal of Rare Earths

journal homepage: http://www.journals.elsevier.com/journal-of-rare-earths

Phase diagrams of permanent magnet alloys: Binary rare earth alloy systems☆

Kaicheng Yang, Jiang Wang*, Qingrong Yao**, Zhao Lu, Maohua Rong, Huaiying Zhou, Guanghui Rao

School of Materials Science and Engineering & Guangxi Key Laboratory of Information Materials, Guilin University of Electronic Technology, Guilin 541004, China

ARTICLE INFO

Article history:
Received 21 July 2018
Received in revised form
27 February 2019
Accepted 28 February 2019
Available online 3 May 2019

Keywords:
Permanent magnets
Rare earth alloys
Phase diagrams
CALPHAD

ABSTRACT

Thermodynamic optimization of the binary rare earth alloy (Ce–La, Ce–Pr, Ce–Nd and La–Nd) systems was performed in this work through the CALPHAD method based on the critical evaluation of all available phase diagram and thermodynamic data reported in the literature. During the thermodynamic modeling, the solution phases including liquid, bcc, fcc and dhcp, were treated as the substitutional solution model. Thermodynamic parameters of the stable phases in the Ce–La, Ce–Pr, Ce–Nd and La–Nd binary systems are obtained finally and would be used directly to develop the thermodynamic database of the multi-component Nd–Fe–B-based alloys, which is indispensable for designing alloy compositions and processes of Nd–Fe–B permanent magnets with highly abundant rare earth metals.

© 2019 Published by Elsevier B.V. on behalf of Chinese Society of Rare Earths.

1. Introduction

Nd–Fe–B permanent magnets with higher coercivity and higher magnetic energy are usually achieved by the addition of heavy rare earth (RE) metals (e.g. Dy, Tb) to 2:14:1-type main-phase.[1,2] The great demand of Nd–Fe–B permanent magnets leads to the massive consumption of expensive rare earth metals (e.g. Nd, Pr, Dy, Tb), while highly abundant rare earth metals (e.g. La, Ce, Y) are overstocked.[3] Many efforts have been made to study the effect of La and/or Ce substitution for Nd and/or Pr on the microstructure and magnetic properties of Nd–Fe–B permanent magnets.[4–7] It was indicated that La substitution reduces the structure stability of the 2:14:1-type main-phase, while the phase segregation of $(Nd_xCe_{1-x})_2Fe_{14}B$ alloys results from the difference of Ce and Nd valence.[4,5] The coercivity of Nd–La–Fe–B magnets decreases due to the inhomogeneous La/Nd distribution in the 2:14:1-type main-phase grains, while the remanence still maintains at a high level.[6] To understand better the phase formation and phase transition of Nd–Fe–B permanent magnets with highly abundant rare earth metals (e.g. La, Ce, Y), phase equilibria and thermodynamic information of the multi-component Nd–Fe–B-based alloys are indispensable.

As key sub-binary systems of multi-component RE–Nd–Fe–B (RE = La, Ce, Pr) alloy system, phase equilibria and thermodynamic properties of the Ce–La, Ce–Pr, Ce–Nd and La–Nd binary systems are first required. Phase diagrams of the Ce–La, Ce–Pr, Ce–Nd and La–Nd binary systems were evaluated early by Gschneidner and Calderwood.[8,9] Due to the lack of experimental information, the phase boundary of these binary phase diagrams was drawn generally by dashed lines. Two assessments of the Ce–La binary system were already reported by Guo et al.[10] and Gong et al.[11] in the thermodynamic calculations of the Ce–La–Mg and Ce–La–O ternary systems, respectively. However, thermodynamic properties of the Ce–La alloys determined by Shoji et al.[12] were not used in their assessments.[10,11] Cacciamani et al.[13] employed the ideal solution model to calculate the Ce–Nd binary phase diagram. The calculated phase boundary of fcc and dhcp phases was not consistent with the experimental data.[14] In addition, the Ce–Nd and La–Nd binary systems have been assessed in our previous

☆ **Foundation item:** Project supported by National Key Research and Development Program of China (2016YFB0700901), National Basic Foundation of China (2014CB643703), National Natural Science Foundation of China (51761008, 51461013), Guangxi Natural Science Foundation (2016GXNSFDA380015, 2016GXNSFGA380001), Guangxi Project of Science and Technology (2017AD23031) and Guangxi Key Laboratory of Information Materials, Guilin University of Electronic Technology, China (171005-Z).

* Corresponding author.
** Corresponding author.
E-mail addresses: wangjiang158@163.com (J. Wang), qingry96@guet.edu.cn (Q.R. Yao).

https://doi.org/10.1016/j.jre.2019.02.003
1002-0721/© 2019 Published by Elsevier B.V. on behalf of Chinese Society of Rare Earths.

work,[15] while the calculated Ce–Pr binary phase diagram was not still reported in the published literature. Therefore, thermodynamic assessment of the Ce–La, Ce–Pr, Ce–Nd and La–Nd binary systems is performed in this work based on the critical review of the reported experimental information using the CALPHAD method,[16] which would be expected to provide reasonable thermodynamic parameters for the development of thermodynamic database of multi-component RE–Nd–Fe–B permanent magnets.

2. Experimental

2.1. Ce–La binary system

Phase equilibria of the Ce–La binary system were studied experimentally by Vogel and Klose[17] with the help of thermal analysis and metallography method. The liquidus and solidus in the Ce–La binary system were re-investigated by Terekhova[18] using the thermal analysis and metallographic method. The two-phase (liquid + bcc) region in the Ce–La binary phase diagram determined by Terekhova[18] was much wider than that measured by Vogel and Klose.[17] On the basis of the experimental results mentioned above, Gschneidner and Calderwood[19] reviewed the Ce–La binary phase diagram. The liquidus and solidus in the Ce–La binary system were drawn by straight lines connecting the melting points and transformation temperatures of pure Ce and La, respectively. There is no available experimental information concerning the solid phase transformation of dhcp and fcc phases.

Shoji et al.[12] determined thermodynamic activities of Ce and La in liquid Ce–La alloys at 1673 K in the whole composition range using mass-spectrometric method. The partial molar Gibbs free energies of Ce and La at 1673 K were calculated from activities values of Ce and La, respectively. The Gibbs free energy of formation of liquid Ce–La alloys at 1673 K was calculated from the partial molar Gibbs free energies of Ce and La. According to the experimental results measured by Shoji et al.,[12] liquid Ce–La alloys show almost ideal mixing behavior due to similar physical and chemical properties of Ce and La. The enthalpy of mixing of liquid $Ce_{50}La_{50}$ alloy at 1673 K was reported by Shoji and Matsui[20] to be 0.70 kJ/mol and was also calculated to be 0.03 kJ/mol using Miedema's model. It should be mentioned that the enthalpy of mixing of liquid $Ce_{50}La_{50}$ alloy reported by Shoji and Matsui[20] was not self-consistent with the experimental data obtained by Shoji et al.[12]

2.2. Ce–Pr binary system

The Ce–Pr binary phase diagram was investigated by Altunbas and Harris[21] by means of resistivity, X-ray diffraction (XRD) and thermal analysis (DTA). There are four solution phases, e.g. liquid, fcc (γ-Ce), bcc (β-Pr, δ-Ce) and dhcp (α-Pr, β-Ce). Based on the measured resistivity curves, pure Pr showed one solid transition (dhcp ↔ bcc), while pure Ce presents two solid transitions (dhcp ↔ fcc and fcc ↔ bcc). Using X-ray diffraction, Altunbas and Harris[21] measured the phase boundary of fcc and dhcp phases. The results indicated that the Ce–Pr alloys in the composition range of 0–65 at% Pr were of fcc structure after annealed at 873 K for 2 h, while the Ce–Pr alloys in the composition range of 74 at% to 100 at% Pr were of dhcp structure. The liquidus and solidus in this binary system were obtained from the DTA measurements. Based on the solid state transitions of the Ce–Pr alloys, a continuous solid solution was formed through fcc and dhcp phases, which is impossible according to the phase rule. In the review of the Ce–Pr binary phase diagram, Gschneidner and Calderwood[22] modified the unreasonable single-phase region of continuous solid solution phase formed from fcc and dhcp phases reported by Altunbas and Harris[21] through the eutectoid reaction (bcc ↔ fcc + dhcp) at about 1023 K.

2.3. Ce–Nd binary system

Speight et al.[14] measured the lattice parameter of binary Ce–Nd alloys after rapidly cooled from 873 K by XRD. Based on the XRD results, the phase boundary of fcc and dhcp phases was extrapolated by linear interpolation. It is indicated that the Ce–Nd alloys in the composition range of 0–30 at% Nd have fcc structure, while the Ce–Nd alloys in the composition range of 40 at% to 100 at% Nd showed dhcp structure. In the review of the Ce–Nd binary phase diagram, Gschneidner and Calderwood[23] used the dashed lines to draw the phase boundary of the Ce–Nd binary phase diagram due to the lack of corresponding experimental information. Keiser[24] investigated the composition of phase boundary of the two-phase (dhcp + fcc) region in the Ce–Nd binary system using the Ce–Nd diffusion couples annealed at 698, 823 and 923 K for 96 and 20 h, respectively. The experimental results[25] showed the two-phase (dhcp + fcc) region was wide and thus is considerably different from the experimental results measured by Speight et al.[14] It should be noted that the Ce–Nd diffusion couples annealed at low temperatures for limited times are insufficient to obtain the reliable information of phase boundary in the diffusion couple experiments.[24]

2.4. La–Nd binary system

The predicted La–Nd binary phase diagram was shown by Gschneidner and Calderwood.[25] There are four solution phases, e.g. liquid, fcc (β-La), bcc (γ-La, β-Nd) and dhcp (α-La, α-Nd). The liquidus and solidus were drawn by dashed lines due to the lack of reliable experimental information. Gschneidner and Calderwood[25] suggested that the bcc solution phase decomposes through the eutectoid reaction (bcc ↔ fcc + dhcp) at about 1098 K. Wu et al.[26] determined that the solubilities of Nd in β-La and La in α-Nd at 773 K were about 35 at% Nd and 60 at% La, respectively. Zhang et al.[27] reported that the solubility of Nd in β-La at 773 K was about 5 at% Nd, while that of La in α-Nd was about 90 at% La, which is considerably different from the experimental results by Wu et al.[26]

Shoji and Matsui[28] determined the activities of Nd in liquid La–Nd alloys at 1673 K in the whole composition range using mass-spectrometry method. Activities of La were calculated using the Gibbs–Duhem equation. The partial molar Gibbs energies of La and Nd at 1673 K were calculated from activities values of La and Nd, respectively. The Gibbs free energy of formation in liquid La–Nd alloys at 1673 K was obtained from the partial molar Gibbs energies of La and Nd. According to thermodynamic properties of liquid La–Nd alloys obtained by Shoji and Matsui,[28] the liquid phase in the La–Nd binary system showed positive deviation from Raoult's law in the whole composition range.

3. Thermodynamic modeling

Based on the experimental information reviewed above, the solution phases including liquid, fcc, bcc and dhcp, in the Ce–La, Ce–Pr, Ce–Nd and La–Nd binary systems were taken into account in this work. The solution phases (ϕ = liquid, fcc, bcc and dhcp) are treated as the substitutional solution model with the Redlich–Kister formula.[29] The molar Gibbs energy of the solution phase can be expressed as:

$$G_m^\phi = {}^0G_A^\phi x_A + {}^0G_B^\phi x_B + RT(x_A \ln x_A + x_B \ln x_B) + x_A x_B \sum_j {}^{(j)}L_{A,B}^\phi (x_A - x_B)^j \quad (1)$$

Table 1
Thermodynamic parameters of the Ce–Pr and Ce–Nd binary systems in this work.

System	Phase	Thermodynamic parameters	Reference
Ce–Pr	liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ce,Pr}} = 0$	This work
	bcc	${}^{0}L^{\mathrm{bcc}}_{\mathrm{Ce,Pr}} = -300$	This work
	fcc	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Pr}} = 900 - 0.5439T + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Pr}}$	33
		${}^{0}L^{\mathrm{fcc}}_{\mathrm{Ce,Pr}} = -250-0.62T$	This work
		${}^{0}G^{\mathrm{dhcp}}_{\mathrm{Ce}} = -190 + 0.56886T + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}}$	31
	dhcp	${}^{0}L^{\mathrm{dhcp}}_{\mathrm{Ce,Pr}} = -120+0.357T$	This work
Ce–Nd	liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ce,Nd}} = 0$	This work
	bcc	${}^{0}L^{\mathrm{bcc}}_{\mathrm{Ce,Nd}} = 0$	This work
	fcc	${}^{0}G^{\mathrm{fcc}}_{\mathrm{Nd}} = 500 + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}}$	32
		${}^{0}L^{\mathrm{fcc}}_{\mathrm{Ce,Nd}} = 0$	This work
	dhcp	${}^{0}G^{\mathrm{dhcp}}_{\mathrm{Ce}} = -190 + 0.56886T + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}}$	31
		${}^{0}L^{\mathrm{dhcp}}_{\mathrm{Ce,Nd}} = -100$	This work

Table 2
The calculated temperatures of invariant reactions in the Ce–Pr, Ce–Nd and La–Nd binary systems with the literature data.

System	Invariant reaction	Type	Temperature (K)	Reference
Ce–Pr	bcc ↔ fcc + dhcp	eutectoid	995.5	This work
			~1023	23
Ce–Nd	bcc ↔ fcc + dhcp	eutectoid	978.4	This work
			975.8	13
			~973	24
La–Nd	bcc ↔ fcc + dhcp	eutectoid	1046.5	This work
			~1098	22

where ${}^{0}G^{\phi}_{\mathrm{A}}$ and ${}^{0}G^{\phi}_{\mathrm{B}}$ are the Gibbs energy of ϕ phase for the element A and B, and were taken from the compilation of Dinsdale.[30] x_{A} and x_{B} are mole fractions of the element A and B, respectively. R is the gas constant and T is temperature (Kelvin). ${}^{(j)}L^{\phi}_{\mathrm{A,B}}$ is the interaction parameter between A and B atoms for ϕ phase in the A–B binary systems as expressed:

$$ {}^{(j)}L^{\phi}_{\mathrm{A,B}} = a + bT \tag{2} $$

where a and b are optimized in this work based on the experimental data.

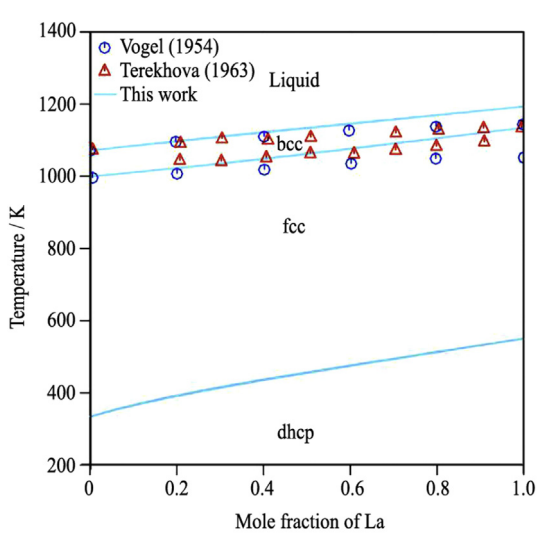

Fig. 1. Calculated Ce–La phase diagram with the experimental data.[17,18]

The Gibbs energy of Ce in the hypothetical dhcp structure and the Gibbs energies of Nd and Pr in the hypothetical fcc structure were not available in the compilation of Dinsdale.[30] These parameters were reported in Refs. 31–33 and were given as:

$$ {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Ce}} = -190 + 0.56886T + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ce}} \tag{3} $$

$$ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Pr}} = 900 - 0.5439T + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Pr}} \tag{4} $$

$$ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Nd}} = 500 + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} \tag{5} $$

4. Results and discussion

All stable phases in the Ce–La and La–Nd binary systems are treated as the ideal solution model in this work. The optimization of

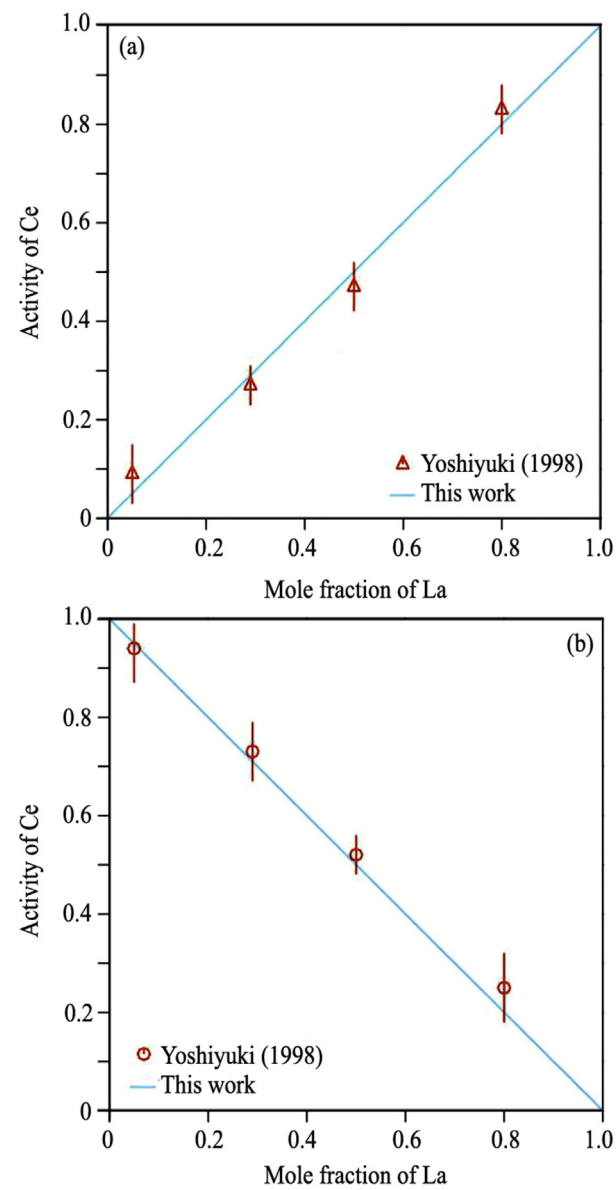

Fig. 2. Calculated activities of La (a) and Ce (b) in liquid Ce–La alloys at 1673 K with the experimental data.[12] (Reference states: liquid-La and liquid-Ce).

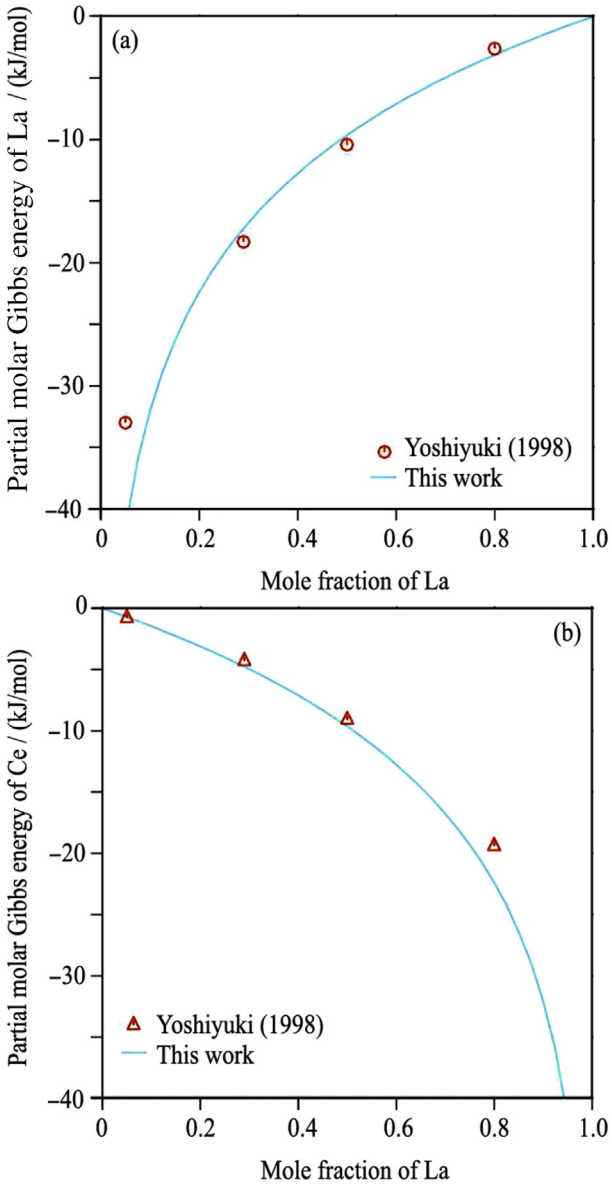

Fig. 3. Calculated partial molar Gibbs free energy of La (a) and Ce (b) at 1673 K with the experimental data.[12] (Reference states: liquid-La and liquid-Ce).

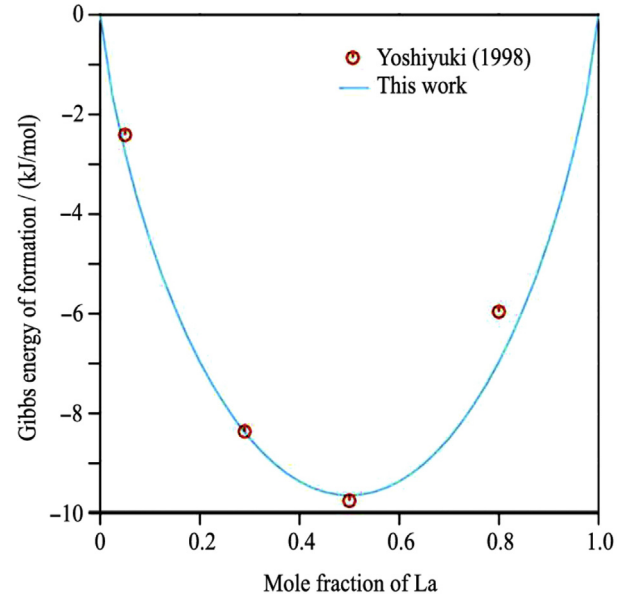

Fig. 4. Calculated Gibbs energy of formation of liquid Ce–La alloys at 1673 K with the experimental data.[12] (Reference states: liquid-La and liquid-Ce).

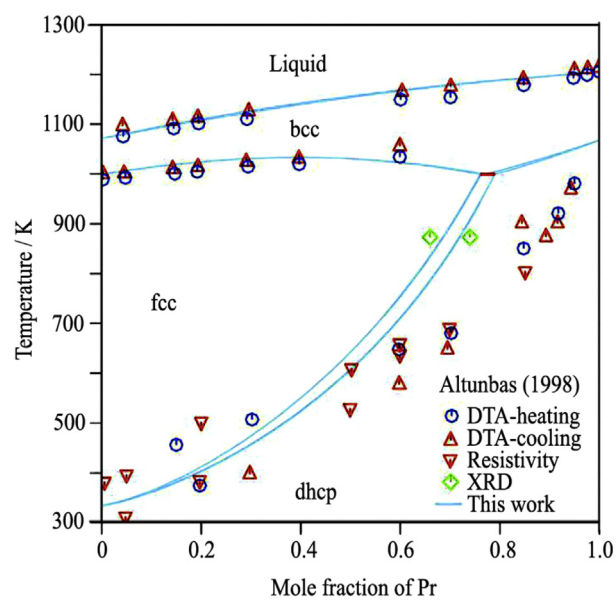

Fig. 5. Calculated Ce–Pr phase diagram with the experimental data.[21]

the Ce–Pr and Ce–Nd binary systems are based on the experimental results determined by Altunbas and Harris[21] and Speight et al.[14] Thermodynamic optimization was carried out by means of the PARROT module in the Thermo-Calc® software.[34] Thermodynamic parameters obtained finally in this work are given in Table 1. Table 2 summarizes the calculated results of invariant reactions in the Ce–Pr, Ce–Nd and La–Nd binary systems with the experimental data. The following section shows the comparison of the calculated results with the experimental data.

4.1. Ce–La binary system

Fig. 1 is the calculated phase diagram of the Ce–La binary system with the experimental results.[17,18] It can be seen that the calculated liquidus and solidus in the La-rich part show some deviation from the experimental results.[17,18] This is due to low purity (97 wt%) of La used in their experiments.[17,18] The calculated phase boundary of fcc and dhcp phases is still necessary to be confirmed by the further experiments. As shown in Fig. 2, the calculated activities of Ce and La in liquid Ce–La alloys agree well with the experimental data[12] within experimental error. The partial molar Gibbs free energies of Ce and La at 1673 K are calculated as given in Fig. 3. Fig. 4 is the comparison of the calculated Gibbs energy of formation of Ce–La alloys at 1673 K and the experimental data.[12] As can be seen, the present calculated partial molar Gibbs free energies and Gibbs energies of formation of Ce and La are in good agreement with the experimental data.[12]

4.2. Ce–Pr binary system

Fig. 5 shows the comparison of the calculated Ce–Pr binary phase diagram with the experimental data determined by Altunbas

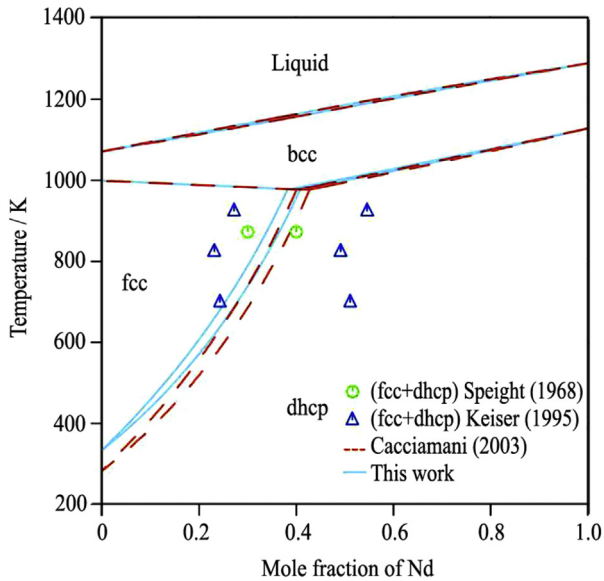

Fig. 6. Calculated Ce–Nd phase diagram with the experimental data[14,24] and the calculated results.[13]

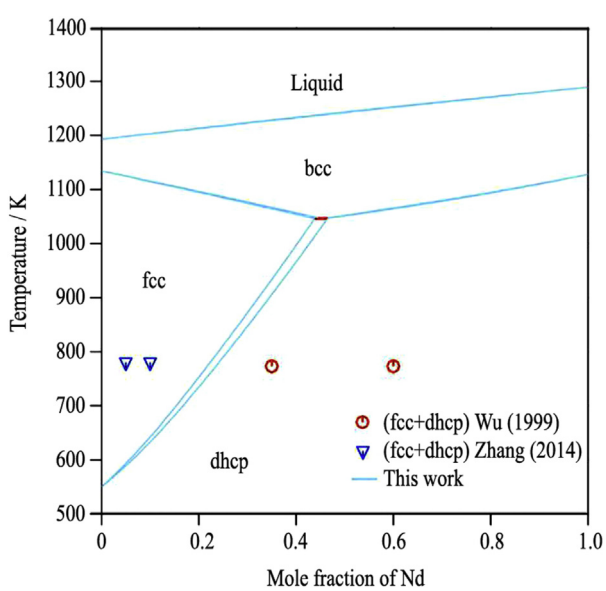

Fig. 7. Calculated La–Nd phase diagram with the experimental data.[26,27]

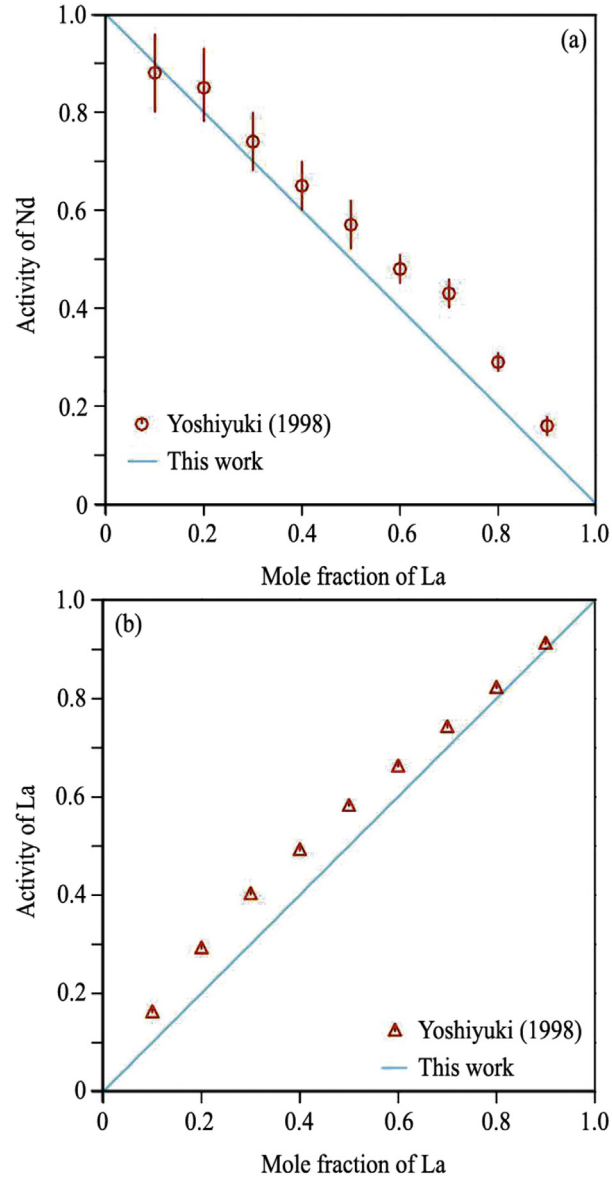

Fig. 8. Calculated activities of Nd (a) and La (b) in liquid La–Nd alloys at 1673 K with the experimental data.[28] (Reference states: liquid-Nd and liquid-La).

and Harris.[21] The calculated phase diagram is in good agreement with most of the experimental data, although there is a slight difference in the phase boundary of fcc and dhcp phases. The calculated results in the present work are still accepted considering the lack of more experimental data. In Table 2, the calculated temperature of the invariant reaction (bcc ↔ fcc + dhcp) is 995.5 K, which is a large deviation from the value (1023 K) reported by Gschneidner and Calderwood.[22]

4.3. Ce–Nd binary system

The comparison of the calculated Ce–Nd binary phase diagram in this work with the experimental data measured by Speight et al.[14] and Keiser[24] and the calculated results by Cacciamani et al.[13] are shown in Fig. 6. The calculated transformation temperature of fcc to dhcp solid solution is not consistent with the calculated results[13] due to the different lattice stability of the dhcp structure of Ce. The calculated phase boundary of fcc and dhcp phases by Cacciamani et al.[13] show slight deviation from the experimental data.[14] In order to fit the experimental data,[14] one interaction parameter ($^{0}L^{\mathrm{dhcp}}_{\mathrm{Ce,Nd}}$) is used in present work. More reasonable agreement is obtained between the calculated and experimental results in the present work. Nevertheless, the phase boundary of fcc and dhcp phases in the Ce–Nd binary system is still necessary to be confirmed by the further experiments.

4.4. La–Nd binary system

Fig. 7 presents the calculated phase diagram of the La–Nd binary system. The calculated phase boundary of fcc and dhcp phases is not consistent with experimental data,[26,27] which is due to the large uncertainty of the experimental results.[26,27] It means that further experiments are needed. The calculated activities of La and

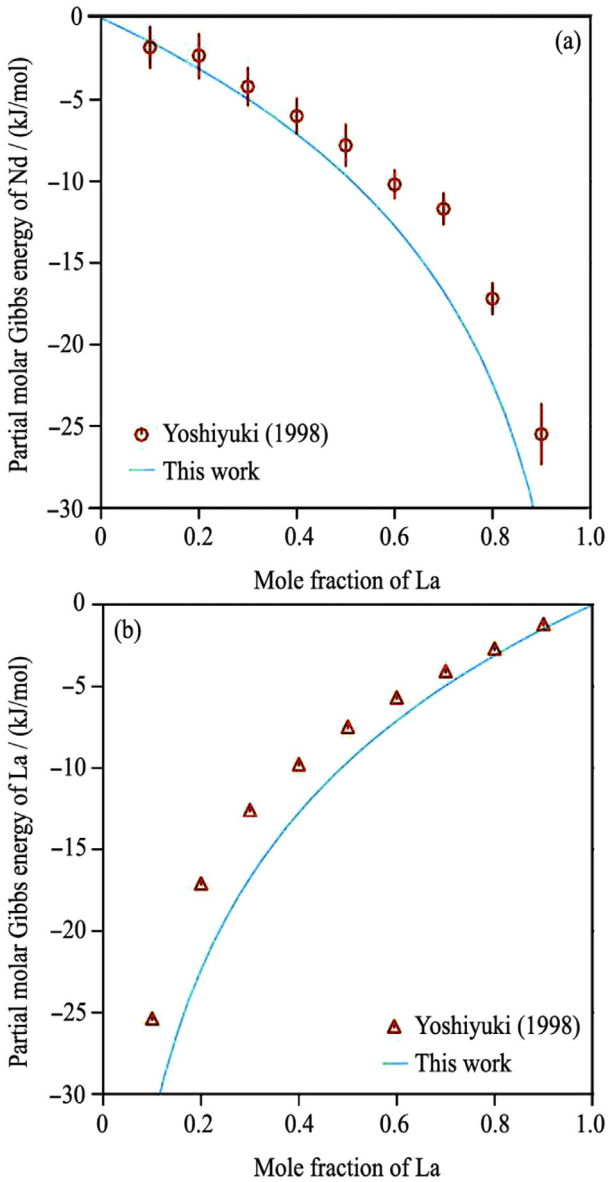

Fig. 9. Calculated partial molar Gibbs free energy of Nd (a) and La (b) at 1673 K with the experimental data.[28] (Reference states: liquid-Nd and liquid-La).

Nd in liquid La–Nd alloys in Fig. 8 are not in good agreement with the experimental data.[28] Fig. 9 is the comparison of the calculated partial molar Gibbs energy of La and Nd at 1673 K with the experimental data determined by Shoji and Matsui.[28] The calculated results are not consistent with the experimental results.[28] Fig. 10 is the calculated Gibbs energy of formation for La–Nd alloys at 1673 K with the experimental data determined by Shoji and Matsui.[28] As can be seen, the calculated results are lower than the experimental results.[28] Thermodynamic properties of liquid La–Nd alloys determined experimentally exhibit positive deviation from the Raoult's law. Due to the similar physical/chemical properties of rare earth metals, it is believed that the liquid La–Nd alloys show to be ideal solution. It indicates that the experimental results of thermodynamic properties of liquid La–Nd alloys reported could be less reliable and should be determined by further experiments. Therefore, any interaction parameters to describe liquid phase are not used in this work.

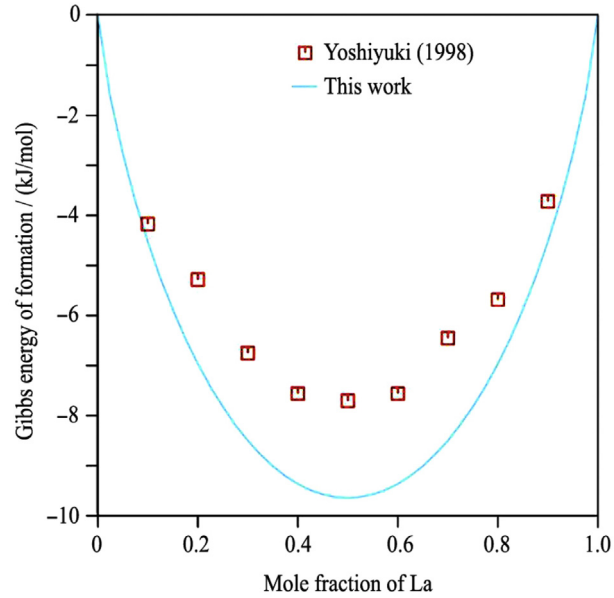

Fig. 10. Calculated Gibbs energy of formation of liquid La–Nd alloys at 1673 K with the experimental data.[28] (Reference states: liquid-Nd and liquid-La).

5. Conclusions

In this work, the Ce–La, Ce–Pr, Ce–Nd and La–Nd binary systems were assessed thermodynamically using the CALPHAD method based on the critical review of the reported experimental information. The calculated results are in good agreements with the experimental data. The comparison of the calculated results with the experimental results suggests that more detailed experiments are still necessary to provide reasonable thermodynamic parameters in the development of thermodynamic database of multi-component Nd–Fe–B magnets.

References

1. Zhou L, Li J, Cheng XH, Liu T, Yu XJ, Li B. Dy gradient and coercivity in grain boundary diffusion processed Nd-Fe-B magnet. *J Rare Earths*. 2017;35:559.
2. Xiang PM, Zhang PY, Li XJ, Gong HL, Wu Q, Jiao WZ, et al. Effect of terbium addition on the coercivity of the sintered NdFeB magnets. *J Rare Earths*. 2010;28:399.
3. Liao XF, Zhao LZ, Zhang JS, Zhong XC, Jiao DL, Liu ZW. Enhanced formation of 2:14:1 phase in La-based rare earth-iron-boron permanent magnetic alloys by Nd substitution. *J Magn Magn Mater*. 2018;464:31.
4. Xu KS, Li HW, Luo Y, Wang LG, Yu DB, Wang ZL, et al. Experimental and computational study on the phase formation and magnetic properties of Ce-La-Fe-B alloys. *J Magn Magn Mater*. 2018;461:100.
5. Pthank AK, Khan M, G Jr K, Mccallum RW, Zhou L, Sun K, et al. Cerium: an unlikely replacement of dysprosium in high performance Nd-Fe-B permanent magnets. *Adv Mater*. 2015;27:2663.
6. Jin JY, Wang Z, Bai GH, Peng BX, Liu YS, Yan M. Microstructure and magnetic properties of core-shell Nd-La-Fe-B sintered magnets. *J. Alloys Compd*. 2018;749:580.
7. Wang L, Wang J, Rong MH, Rao GH, Zhou HY. Effect of wheel speed on phase formation and magnetic properties of $(Nd_{0.4}La_{0.6})$-Fe-B melt-spun ribbons. *J Rare Earths*. 2018;36:1179.
8. Gschneidner KA, Calderwood FW. Use of systematics for the evaluation of rare-earth phase diagrams and crystallographic data. *Bull Alloy Phase Diagram*. 1983;4:129.
9. Gschneidner KA, Calderwood FW. *Intra Rare Earth Binary Alloys: Phase Relationships, Lattice Parameters and Systematics*. In: *Handbook on the Physics and Chemistry of Rare Earths*. vol. 8. Amsterdam: Elsevier Science & Technology; 1986:9 [Chapter 54].
10. Guo CP, Du ZM, Li CR. A thermodynamic description of the Ce-La-Mg system. *Int J Met Res*. 2011;101:1424.
11. Gong WP, Zhang R, Chen ZS. Thermodynamic modelling and applications of Ce-La-O phase diagram. *Trans Nonferrous Met Soc China*. 2011;21:2671.
12. Shoji Y, Matsui T, Nakamura K, Inoue T. Vaporization study on lanthanum-cerium alloys by mass-spectrometric method. *J Nucl Mater*. 1998;254:9.

13. Cacciamani G, Cardinale AM, Borzone G, Ferro R. Thermodynamic modeling and optimization of the Al-Ce-Nd system. *CALPHAD*. 2003;27:227.
14. Speight JD, Harris IR, Raynor GV. Alloys of cerium with neodymium, samarium and terbium, and of praseodymium with terbium. *J Alloys Compd*. 1968;15:317.
15. Yang KC, Xu L, Rong MH, Yao QR, Wang J. Thermodynamic calculation of phase diagrams for the rare earth binary alloys: RE-Nd (RE = La, Ce, Pr, Y) systems. In: Han Y, ed. *Advances in Materials Processing*. Singapore: Springer; 2017:1305.
16. Sundman B, Jansson B, Anderson JO. Computer coupling of phase diagram and thermochemistry. *CALPHAD*. 2011;26:599.
17. Vogel R, Klose H. Über die Zustandsbilder Cer-Lanthan, Lanthan-Antimon und Cer-Indium. *Z Metallkd*. 1954;45:633.
18. Terekhova VF. Alloys of rare earth metals. *Met Sci Heat Treat*. 1963;47:464.
19. Gschneidner KA, Calderwood FW. The Ce-La system. *Bull Alloy Phase Diagram*. 1982;2:445.
20. Shoji Y, Matsui T. Thermodynamic activity and Gibbs energy of mixing of rare earth alloys. *J Mass Spectrom Soc Jpn*. 1999;47:23.
21. Altunbas M, Harris IR. Structural and constitutional studies of some cerium-praseodymium alloys. *J Mater Sci*. 1980;15:693.
22. Gschneidner KA, Calderwood FW. The Ce-Pr system. *Bull Alloy Phase Diagram*. 1982;3:187.
23. Gschneidner KA, Calderwood FW. The Ce-Nd system. *Bull Alloy Phase Diagram*. 1982;3:89.
24. Keiser D. Diffusion in the Ce-Nd system. *Scr Mater*. 1995;33:959.
25. Gschneidner KA, Calderwood FW. The La-Nd system. *Bull Alloy Phase Diagram*. 1982;3:25.
26. Wu SW, Zhang LP, Zeng LM, Zhuang YH. Phase relationship of the La-Nd-Fe at 770 K. *J Alloys Compd*. 1999;291:220.
27. Zhang F, Xu HH, Du Y, Schmid-Fetzer R, Zhou T. Phase equilibria of the Mg-La-Nd system at 500 °C. *J Alloys Compd*. 2014;585:384.
28. Shoji Y, Matsui T. Vaporization study on lanthanum-neodymium alloys by mass-spectrometry. *J Nucl Mater*. 1999;273:310.
29. Redlich O, Kister AT. Algebraic representation of thermodynamic properties and the classification of solutions. *J Ind Eng Chem*. 1948;40:345.
30. Dinsdale AT. SGTE data for pure elements. *CALPHAD*. 1991;15:317.
31. Meng FG, Liu LB, Liu HS, Jin ZP. Thermodynamic assessment of the Ce-Y system. *CALPHAD*. 2006;30:323.
32. Wang Y, Curtarolo S, Jiang C, Arroyave R, Wang T, Ceder G, et al. Ab initio lattice stability in comparison with CALPHAD lattice stability. *CALPHAD*. 2004;28:79.
33. Jin L, Kang YB, Chartrand P, Fuerst CD. Thermodynamic evaluation and optimization of Al-La, Al-Ce, Al-Pr, Al-Nd and Al-Sm systems using the modified quasichemical mode for liquids. *CALPHAD*. 2011;35:30.
34. Sundman B, Jansson B, Andersson JO. The Thermo-Calc databank system. *CALPHAD*. 1985;9:153.